# Saptarshi Mandal


Fourth year undergraduate,
Department of Mathematics,
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EDUCATION

**Indian Institute of Technology Kanpur**
BS in Mathematics & Scientific Computing (CPI: 8.9 / 10.0) 2023 - 2027 (ongoing)
Minor in Theory of Computation
MS in Mathematics 2027 - 2028 (upcoming)

INTERESTS

Set Theory, Model Theory & Logic and its applications to other areas of mathematics
Geometric Measure Theory, Fourier & Harmonic Analysis, General Topology

PRESENTATIONS

"Hat Guessing with Proper Colorings", **Joint Mathematics Meetings 2026**, Washington DC, USA, January 2026.

PREPRINTS

"Hat Guessing with Proper Colorings", **Saptarshi Mandal**, Anurag Ramachandran, Michael Kreiger, Peter Bentley, 2025. [arXiv]

RESEARCH

**Sticky Kakeya Conjecture** | ICTS-TIFR Summer Research May 2026 - Jul 2026
*Dr. Trishen Gunaratnam (TIFR Mumbai)*
• Studied the *Wang-Zahl (2022)* proof of the sticky Kakeya conjecture in detail, learning geometric measure theory alongside; understood the full argument except the proof of 5.2
• Delivered a 20-minute presentation on the structural properties: coarse self-similarity, plaininess, graininess; wrote a detailed report expanding on several proofs from the paper.

**Hat Guessing Number of Graphs** | Polymath Jr., 2025 May 2025 - Jul 2025
*Dr. Anurag Bishnoi (TU Delft)*
• Proved that the proper hat guessing number of the complete graph $K_n$ is $2n - 1$.
• Assisted in the proofs of some general bounds on the proper hat guessing number.

**Abstract Harmonic Analysis** | SURGE 2025, IIT Kanpur May 2025 - July 2025
*Dr. Arijit Ganguly (IIT Kanpur)*
• Studied basic Fourier theory for Hilbert Spaces and LCA Groups.
• Reconstructed proofs of Fubini's Theorem and the General Plancherel Theorem.

PROJECTS

**Topics in set theory & model theory** | Undergraduate Project Jul 2026 - present
*Dr. Ashutosh Kumar (IIT Kanpur)*
• Studying *Keisler*'s survey on ultraproducts to build a strong background in model theory.
• Working towards *Malliaris-Shelah (2014)* proof of $\mathfrak{p} = \mathfrak{t}$ over the coming months.

**Topics in number theory** | Undergraduate Project Jul 2026 - present
*Dr. Somnath Jha (IIT Kanpur)*
• Reading *Serre*'s *A Course in Arithmetic* with the goal of building a working knowledge of analytic number theory.

**Function with a divergent Fourier Series** | *course presentation*
**Riesz-Thorin Interpolation Theorem** | *course presentation*
**Prime factorization of ideals in a Dedekind domain** | *course presentation*

**Type theory & Functional Programming** May 2024 - July 2024
*Stamatics Society, IIT Kanpur*
• Explored fundamental type theory concepts, like type inference and polymorphism, algebraic data types, parametric polymorphism, monads and concurrency.
• Implemented the above in OCaml.

ACHIEVEMENTS

ICTS - **S.N. Bhatt Memorial Excellence Fellowship Program** 2026
**Academic Excellence Award**, IIT Kanpur (top 10% students) 2024
**Rank 2848**, JEE Advanced (out of 150,000 candidates) 2023
**Rank 7449**, JEE Mains (out of 1.4 million candidates) 2023
**Rank 66**, WBJEE (out of 120,000 candidates) 2023

COURSEWORK

**Logic:** Set Theory & Logic; Data Structures & Algorithms; Fundamentals of Computing; Topics in Model Theory*; Computability Theory*; Theory of Computation*

**Analysis:** Real Analysis, Complex Analysis, Measure Theory, Ordinary Differential Equations, Partial Differential Equations, Euclidean Fourier Analysis, Functional Analysis*

**Algebra:** Linear Algebra, Abstract Algebra, Field & Galois Theory, Algebraic Topology, Arithmetic on Quadratic Fields

**Others:** Several Variable Calculus & Differential Geometry, Probability Theory, Numerical Analysis & Scientific Computing I & II, Digital Electronics

SKILLS

**Programming:** C, C++, Python, Rust, Bash, Prolog, OCaml
**Utilities:** Git, Linux, Latex, MATLAB, SageMath

POSITIONS

**Coordinator**, Stamatics Society, IIT Kanpur Oct 2025 - present
**General Executive**, Stamatics Society, IIT Kanpur Nov 2024 - Oct 2025
**Blog Writer**, Stamatics Society, IIT Kanpur Nov 2024 - Oct 2025
**Student Guide**, Institute Counselling Service, IIT Kanpur Aug 2024 - present
**Academic Mentor**, Institute Counselling Service, IIT Kanpur Aug 2024 - present